Question	Answer	Marks	Partial Marks
5(a)	–1, –0.375, 3	**3**	**B1** for each
5(b)	Correct graph	**4**	**B3FT** for 8 or 9 correct points or **B2FT** for 6 or 7 correct points or **B1FT** for 4 or 5 correct points
5(c)	$y = 2 - x$ ruled correctly AND –0.45 to –0.35 1 2.35 to 2.45	**4**	**B2** for $y = 2 - x$ ruled or **B1** for [y =] $2 - x$ soi or $y = k - x$ ruled or $y = kx + 2$ ruled, but not $y = 2$ **B2** for all three values or **B1** for any two values
6(a)(i)	4.5, $4\frac{1}{2}$ or $\frac{9}{2}$	**3**	**M1** for $8x - 12 = 24$ or $2x - 3 = 6$ **M1** for reaching $ax = b$ correctly FT *their* first step
6(a)(ii)	$x > -\frac{4}{3}$ or $x > -1\frac{1}{3}$ final answer	**2**	**M1** for $6x > 6 - 14$ or $x + \frac{14}{6} > 1$
6(b)	[y =] $\sqrt[3]{\frac{2x^3 - V}{3}}$ oe final answer	**3**	**M1** for isolating term in y **M1** for division by 3 or FT *their* first step **M1** for cube root or FT *their* previous step to the final answer
6(c)	$4n^2 - 20n + 12$	**M2**	**B1** for $4n^2 - 10n - 10n + 25$
	$4(n^2 - 5n + 3)$ or correct explanation linked to expression	**A1**	with no errors seen e.g. 4, [–]20 and 12 are all multiples of 4 or divides each term or each coefficient by 4
6(d)(i)	$p = -3$ and $q = 23$	**3**	**B2** for $23 - 2(x - 3)^2$ OR **M1** for [q] $- 2x^2 - 4px - 2p^2$ or $-2(x - 3)^2$ seen **B1** for either $p = -3$ or $q = 23$ or **FT** $q = 5 + 2(\textit{their } p)^2$
6(d)(ii)	(3, 23)	**1**	**FT** *their* (d)(i)
6(e)	69	**2**	**M1** for figs 13^2 oe
7(a)(i)	$\frac{16\pi}{3}$ or $5\frac{1}{3}\pi$ final answer	**2**	**M1** for $\frac{1}{2} \times \frac{4}{3}\pi \times 2^3$ oe

Question	Answer	Marks	Partial Marks
7(a)(ii)	2.4[0]	**4**	**B3** for answer in range 2.396… to 2.40… OR **M3** for *their* $\frac{16\pi}{3} + \pi \times 2^2 \times 5.2 + \frac{1}{3}\pi \times 2^2 \times h = \frac{88\pi}{3}$ oe or **M2** for $\frac{88\pi}{3}$ – *their* $\frac{16\pi}{3} - \pi \times 2^2 \times 5.2$ oe or **M1** for $\pi \times 2^2 \times 5.2$ oe or $\frac{1}{3}\pi \times 2^2 \times h$ oe soi
7(a)(iii)	1 hour 38 min or 1 hour 37.8 min to 1 hour 37.9… min	**3**	**B2** for 1.63[2…] or 98 [mins] or 97.8 to 97.9…] or **M1** for $\frac{\frac{88\pi}{3} \times 620}{35000}$ [× 60] oe
7(b)	8.5[0] or 8.496 to 8.497	**4**	**M3** for [r=] $\sqrt{\frac{65}{\frac{140}{360}\pi - \frac{1}{2}\sin 140}}$ oe or **M2** for $\frac{140}{360}\pi \times r^2 - \frac{1}{2}r^2 \times \sin 140$ [=65] oe or **M1** for either area expression seen
8(a)(i)	$\frac{12}{x}$ or $12 \div x$ final answer	**1**	
8(a)(ii)	$\frac{12}{x-4}$ – *their* $\frac{12}{x} = 1.5$oe	**M1**	Accept 3 or more term equivalents
	$12x - 12(x-4) = 1.5x(x-4)$ or $\frac{12x - 12(x-4)}{x(x-4)}$ [= 1.5]	**M1**	Correctly clearing fractions, or correctly collecting into a 'single fraction' FT *their* expression dep on two fractions both with algebraic denominators
	$12x - 12x + 48 = 1.5x^2 - 6x$	**M1**	Correctly multiplying *their* two sets of brackets FT *their* expression dep on two fractions both with algebraic denominators or first M1 given
	[$1.5x^2 - 6x - 48 = 0$] $x^2 - 4x - 32 = 0$	**A1**	One further step either 3 term equation or division throughout by 1.5 leading to solution With no errors or omissions seen, dep on **M3**